

Enze Shi

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EDUCATION

Duke University

Master of Science, Electrical & Computer Engineering, **Software Engineering**

Durham, NC

Aug 2022 - Dec 2023

Pennsylvania State University - University Park

Bachelor of Science, **Computer Science**; Minor in Mathematics

State College, PA

Aug 2018 - May 2022

SKILLS

- **Programming Languages:** Python, Java, Swift, C, C++, JavaScript, Kotlin, Golang, Shell
- **Databases:** MySQL, MongoDB, Data Warehousing, Data Integration, ETL (Extract, Transform, Load), SAP Data Services, Smart Data Integration (SDI), Smart Data Access (SDA), Real-time Analytics
- **Frameworks and Tools:** Django, React.js, React Native, Docker, Kubernetes, OpenShift, AWS (EC2, EKS), Serverless Functions, RESTful APIs, Microservices, Unit Testing (Mockito), Grafana, CI/CD
- **iOS Development:** iOS SDK, UIKit, Core Data, Custom UI, Multithreading, Memory Management
- **Software Development:** Distributed Systems, Full SDLC, Data Structures, Algorithms, Large-Scale Application Architecture, Cross-functional Collaboration, Agile, Scrum, Linux

WORK EXPERIENCE

TikTok Inc.

Bellevue, WA

Software Engineer (Site Reliability Engineer - Recommendation Infra)

Jan 2024 - Present

- Engineered a resource management system for core Recommendation Infrastructure to ensure stability in high-concurrency scenarios using **Python, TypeScript**.
- Designed and implemented a Resource Quota subsystem using **Django, React.js**, automating budgeting, business inventory, reducing budget overruns by 29%.
- Spearheaded enhancements to the Resource Allocation Platform by automating requests and approvals, cutting response time by 40% and reducing operational costs.
- Developed real-time monitoring and visualization dashboards using **D3.js** and **Grafana**, providing comprehensive insights into resource usage and enabling instant identification of shortages within seconds.
- Mentored interns and led cross-team collaboration for the handoff of recommendation system resource management, ensuring smooth transitions and improving workflow efficiency.

INTERNSHIP

Duke University

Durham, NC

SSI Systems Engineer Intern - Kubernetes Operations (Python, Kubernetes, Prometheus, Grafana)

June 2023 – Dec 2023

- Successfully migrated Linux@Duke Mirror to new Kubernetes clusters with zero service interruptions and minimal impact on user access, adapting existing container definitions for a seamless transition.
- Enhanced Kubernetes monitoring by customizing Prometheus queries and alerts, reducing false positives by 47% and improving critical issue identification for increased system reliability and reduced downtime.
- Explored and evaluated application deployment strategies, such as canary releases, blue-green deployments, and rolling updates, in Kubernetes to facilitate smooth updates and minimize user impact.

Hirebeat Inc.

New York City, NY

Software Development Engineer Intern (Python, JavaScript, Django, React.js)

May 2020 – Sep 2020

- Delivered client solutions to job-seekers and recruiters by developing a web platform using Python, JavaScript, HTML, CSS, and PostgreSQL database.
- Constructed a highly reusable and cross-browser compatible RESTful client entry system in JavaScript using React, Redux, and Bootstrap4 frameworks, boosted the new user increasing rate by 30%.
- Enhanced functionality by implementing new features, such as billing, client profile management, and email verification, in Python using the Django framework with the model-view-controller (MVC) architecture.
- Improved the interview training experience by creating and overseeing a comprehensive PostgreSQL and AWS-powered database, currently hosting over 5000 recorded videos/audios, catering to a vibrant community of over 1000 active users.

PROJECTS

Campus Map (Swift, MapKit, UIKit, CoreLocation, CloudKit, CoreData)

- Developed a multi-platform project using Swift applied with Model-View-ViewModel (MVVM) architecture to enable users to familiarize themselves with the campus layout.
- Implemented features to enable users to search for a specific location on the campus map and navigate to it in estimated time.
- Built a cloud-based database to store user's preferences and statistics.
- Implemented a todo list feature in priority hierarchy to help users to organize their visit.